

In this lab, I created simulated network traffic data using hourly timestamps for one year. The data included daily traffic patterns, weekly patterns, seasonal changes, a long-term trend, random noise, and traffic spikes.

After the data was generated, I explored it by using graphs and summary statistics. What I found was that the traffic was higher during the day and lower on weekends. This shows that network usage was following a repeating time-based pattern.

I then split the data into training and testing sets. The training set was used to build the models and contained 80% of the data and was used to teach each model the patterns in the network traffic. The testing set was used to evaluate how well the models predicted future traffic with the remaining 20% of the data.

The three models I tested with were, ARIMA, Linear Regression, and LSTM. ARIMA used past traffic values to forecast future traffic. Linear Regression used engineered features created from the original data instead of relying only on previous traffic values. This included hour, weekday, lag values, and moving averages. Finally, the LSTM model is a type of neural network designed for sequential data. Instead of using manually engineered features, it learns patterns directly from sequences of previous traffic values.

After comparing the models using MSE, MAE, and RMSE, these metrics show me how close the predicted network traffic values are to the actual traffic values in the testing dataset. MSE calculates the average of the squared prediction errors which gives more weight to larger errors. MAE measures the average absolute difference between the predicted and actual values. This makes it easy to understand our average prediction error. RMSE is the square root of MSE and expresses the error in the same units as the traffic data.